MILITARY SPECIFICATION

T-BOLTS, 180,000 PSI, TYPE 431 STAINLESS STEEL, MANUFACTURING PROCESS REQUIREMENTS.

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification presents material and processing requirements for type 431 stainless steel T-bolts.

2. APPLICABLE SPECIFICATION, OTHER PUBLICATIONS AND DRAWINGS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this publication to the extent specified herein.

2.1.1 Specifications

Federal

PPP-B-636 Boxes, Fiberboard
PPP-B-566 Boxes, Folding, Paperboard
PPP-B-676 Boxes, Set up, Paperboard

Military

MIL-B-6812 Bolts, Aircraft
MIL-H-6875 Heat Treatment of Steels (Aircraft), Process for
MIL-I-6868 Inspection Process, Magnetic Particle
MIL-S-8879 Screws Threads, Controlled Radius Root with Increased Minor Diameter
MIL-S-18732 Steel Bars, Billets, Forgings, Tubing, (431) Special Quality
MIL-P-116 Preservation, Methods of

2.1.2 Standards

Federal

Fed. Test Method Std. No. 151 Metals; Test Methods

Military

MIL-STD-10 Surface Roughness, Waviness, and Lay
MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for shipment and storage
MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

AMS 2640F - Magnetic Particle Inspection

(Copies of Aeronautical Materials Specifications may be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York 17, New York.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C.)

NATIONAL AIRCRAFT STANDARDS COMMITTEE

NAS 1069 (Rev. 30 June 1959) - Tension Fatigue Test Procedure for Aeronautical Fasteners

(Copies of NAS specification may be obtained from the National Standards Association, 1321 14th Street, NW, Washington, D.C. 20005.)

3. REQUIREMENTS

3.1 Materials - Material shall conform to the requirements of MIL-S-18732 except that free ferrite not in excess of 10 percent will be acceptable. The presence of aligned carbides in the microstructure is not acceptable. Materials shall be capable of being cold headed without the formation of discontinuities or defects.

3.1.1 Heat Treatment - Bolts shall be heat treated in accordance with MIL-H-6875 and to the HT 200 condition except that the austenitizing time shall be 1 hour and refrigeration will not be required.

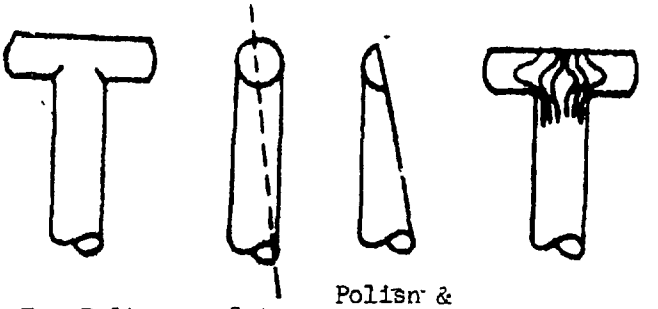
3.1.2 Surface contaminants - The hardened blanks shall be free from decarburization, carburization, and surface contamination or penetration by oxygen, nitrogen or other undesirable elements prior to thread rolling. The processes employed to avoid or remove surface oxides or other contaminants shall not effect the resistance to corrosion, stress corrosion or fatigue.

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3.2 Design and construction - The configuration and dimensions of the bolt shall conform to the applicable drawings and standards, and to the requirements of this specification.

3.3 Heads -

3 3.1 Method of forming - Bolt heads shall be warm or cold forged. A fillet of not less than 0.030 inch radius shall be provided between head and shank surfaces. Flow lines in the finished part when examined as indicated by 4.5.3 shall follow the contour of the head, head to shank, fillet, and bearing surface as illustrated by Figure 1. Slight cutting of the flow lines by the oxide removal process is permissible but in no case will re-entrant angles greater than 35 degrees be acceptable in the fillet area.



Figure

3.4 Shot peening of fillet - The transition zone between head and shank shall be shot peened, employing automatic equipment and glass beads, at an Almen intensity of .008. Adjacent areas may or may not be peened except that threaded areas shall be masked or otherwise protected to avoid peening.

3.5 Straightness of shank - The straightness of the bolt shank shall be within the values specified in Table I.

Table I

Straightness of Shank

Bolt Size	Deviation of Bolt Shank from Plate INCH/INCH of Bolt Length (max)
#10, 1/4, 5/16	.0030
3/8, 7/16	.0025
1/2 and larger	.0020

3.5.1 Squareness of shank and threads with head - The bearing surfaces of bolt heads shall be $90^{\circ} \pm 2^{\circ}$ with the axis of the shank, and of the threaded portion of the shank.

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3.5.2 Surfaces of head and shank -

3.5.2.1 Surfaces of head and shank shall be smooth, with a roughness height rating less than 125 in areas not shot peened, and in all areas prior to shot peening. Surfaces shall be free from longitudinally oriented scratches, notches, folds, scale, or other discontinuities of a depth greater than 1 percent of the shank diameter; and shall be free from all discontinuities having transverse orientation with regard to the centerline of the shank or head, respectively, or extending across threads. Discontinuities and magnetic particle indications exceeding one inch in length are rejectable.

3.6 Threads - The threads shall be right hand. Thread dimensions, form and contour shall conform to specification MIL-S-8879.

3.6.1 Forming - Threads shall be fully formed by a rolling process subsequent to hardening by heat treatment. Flow lines shall be continuous, shall follow the thread contour, and shall be of maximum density at the roots of the threads as illustrated by Figure 2.



Figure 2
Flow Lines
Rolled Thread

3.6.2 Incomplete threads - Incomplete threads, within the following limits, are permitted between the last full thread and the unthreaded shank and at the chamfered end of the bolt. Thread run-out adjacent to the grip shall be a minimum of one pitch length and maximum of two pitch lengths. The run-out shall fair into the shank, thereby eliminating abrupt changes in cross-sectional area. Bottom and sides of threads contained in run-out may deviate from true form but shall be smooth and free from tool marks. The point end of the bolt shall, except for identification, be flat and chamfered. The chamfer to be approximately one pitch. Threads adjacent to the chamfer may be incomplete at pitch diameter and major diameter; but shall be completed and within tolerance at a distance of two pitch lengths from the point end of the bolt.

3.6.3 Thread surfaces - Surfaces of threads shall be smooth and free from notches, slivers, folds, laps, and scale. Surface finish of thread and thread root areas shall not exceed 32 RHR.

3.6.4 Passivity - Surfaces shall be passive when tested in accordance with 4.5.6.

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3.7 Properties -

3.7.1 Strength in tension - Bolts shall be capable of supporting the Table II loads without breaking.

Table II

Tensile Strength	
Thread Size	Load Pounds
#10-32	4,000
1/4-28	7,200
5/16-24	11,400
3/8-24	17,000
7/16-20	23,000
1/2-20	30,700

3.7.1.1 Elongation - The length of bolts shall be measured before and after applying the Table II loads from flats ground on both ends. To meet the required (9% minimum) elongation, the exposed threads, gage length, and minimum increase of length shall be as listed in Table III.

Table III

Size	Stress Area	Equiv. Diam.	Exposed Thd. 5D	4D	Min. Elong.
10-32	.0223	.1686	.843	.674	.061
1/4-28	.0400	.2256	1.128	.902	.081
5/16-24	.0634	.2841	1.421	1.136	.102
3/8-24	.0943	.3465	1.733	1.386	.125
7/16-20	.1278	.4035	2.018	1.64	145
1/2-20	.1705	4659	2.330	1.864	.168

3.7.2 Fatigue - Bolts shall be capable of withstanding 65,000 cycles average life, and 45,000 cycles minimum when cyclically loaded as indicated by Table IV and paragraph 4.5.9. Cycles in excess of 100,000 shall not be used in the computation of averages.

Table IV

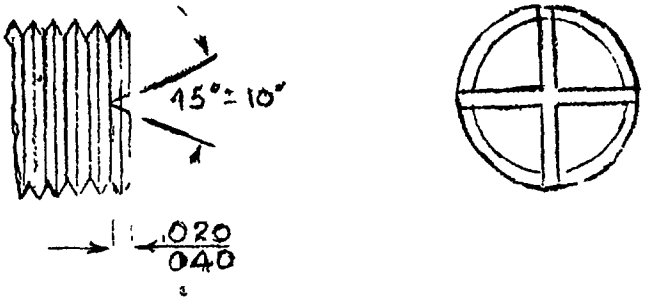
	Cyclic Load Levels	
Nominal bolt diameter (inches)	Minimum load - lbs. ± 2 percent	Maximum load - lbs. ± 2 percent
#10-32	146	1,460
1/4-28	271	2,710
5/16-24	435	4,350
3/8-24	671	6,710
7/16-20	905	9,050
1/2-20	1,230	12,300

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3.8 Magnetic quality - There shall be no transverse magnetic particle indications. Any longitudinal indications shall be light, not more than 5 in number on any one item, with none more than 1 inch in length

3.9 Identification of product - Each bolt shall be identified by the manufacturers trademark impression stamped in an unstressed area of the crossbar or head, plus crossed grooves across the threaded end of the shank. The bolt ends shall be grooved parallel ($\leq 3^{\circ}$) and perpendicular to the axis of the T-head. Grooves shall be within $\leq$ .010 inch of the center of the tip. Groove dimensions shall comply with Figure

FIGURE 3



4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary toassure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection - The examination and testing of the bolts shall be classified as quality conformance inspection.

4.3 Lot - A lot shall consist of finished bolts of one configuration and size, from one heat of steel, heat treated in the same manner and produced as one continuous run or order or part thereof.

4.4 Sampling - Sampling for threads, finish dimensions, and surface roughness shall be at random in accordance with MIL-STD-105, Inspection Level II, for the following AQL's applying to the corresponding class of characteristics:

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Classification	AQL	Dimensional Characteristic
Major	1 0%	Thread size and form Shank diameter Imperfect threads Grip length Straightness of Shank Squareness of shank & threads to head Surface roughness Identification
Minor	4.0%	Over-all length Head width Head height Chamfer of thread end Burrs and tool marks

4.4.1 Sampling for mechanical and metallurgical tests will be as follows:

From each production run by mill heat and heat treat furnace lot, the number of samples shall be in accordance with Table V.

Table V

Test	Sample size per lot
Tensile strength	5
Elongation	5
Bend Test	5
Fatigue	5
Decarburization	2
Surface Contamination and Micro Examination	2

4.4.2 Sampling for metallurgical examination of the microstructure shall be made from each production run. One sample will be required from each lot.

4 5 Methods -

4.5.1 Examination of product - Samples selected in accordance with 4.4 shall be examined for compliance with configuration and dimensions, identification marking, and preparation for delivery.

4.5.2 Thread elements - The configuration and dimensions of thread elements shall be inspected for compliance with MIL-B-8879 requirements

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4.5.3 Metallurgical examination - A section cut as indicated in the Figure 3 shall be polished and suitably etched for examination of the microstructure Examination shall be at magnifications not less than 200x, for compliance with the requirements of 3.1 and 3.3.1.

Subsequent to examination of the microstructure the section shall be further etched to reveal the forging grain flow lines, which shall closely follow the surface contours in the head-to-shank fillet area to the extent indicated by figure

4.5.4 Surface roughness - Surface roughness shall be determined in accordance with the provisions of MIL-STD-10.

4.5.5 Decarburization - An inclined surface extending from an original surface on the shank of test items shall be ground to extend to a depth of .015" or more at one end. Rockwell superficial 15 N hardness tests in accordance with Method 243 of FED STD NO. 151 shall be made on or near an original surface and at a depth below which there is no further change in hardness. Test surface and supporting surfaces shall be parallel within 7°.

4.5.6 Passivation - Passivation shall be checked by swabbing the parts with copper sulfate solution, keeping the surface wet for a period of six minutes.

After this exposure, wipe the surface dry and examine for areas of deposited metallic copper. A copper deposit will indicate the presence of metallic iron.

COPPER SULFATE SOLUTION

$CuSo_4 \cdot 5\ H_2O$	8 grams
Distilled Water	500 ml
H_2SO_4 (Sp. Gr. 1 84)	3 ml

4.5.7 Magnetic inspection - Inspection shall be in accordance with Specification MIL-I-6868 or AMS 2640. Each bolt shall be inspected by both the longitudinal and circular methods

4 5.8 Tensile tests - Tensile tests of the head-to-threaded area shall be conducted in accordance with the applicable requirements of FED TEST METHOD STD. NO. 151, METHOD 211. Appropriate saddle type loading fixtures shall be employed. After applying the Table II proof loads, which should not cause failure, the loads shall be increased until failure occurs. After failure the elongation of the bolt shall be determined for compliance with Table III.

4.5.9 Fatigue - The fatigue testing machines specimen holding fixtures (modified to accommodate the head configuration), nuts, and test setup shall conform to NAS 1069. The fatigue test shall be conducted at room temperature. The fatigue loading shall be tension-tension and shall conform to Table IV. Specimens shall comply with the requirements of 3.7.2.

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4.10 Rejection - Failure of a quality conformance inspection test specimen to meet specified requirements shall be a cause for rejection of the lot represented. Bolts from a rejected lot may be resubmitted for acceptance provided the lot has been completely screened and all defectives removed by the supplier.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be in accordance with Level A or C of MIL-STD-794, as specified in the contract or order (see 6.2). Only bolts of one type, size and part number shall be packaged in the unit container.

5.1.1 Level A - Level A shall conform to Method IC of MIL-P-116, utilizing variety 2 of PPP-B-566 or PPP-B-676.

5.1.2 Level C - Level C shall be the minimum required to provide adequate protection during domestic shipment from the source to the first receiving activity for immediate use.

5.2 Packing - Packing shall conform to Level B or C of MIL-STD-794, as specified in the contract or order (see 6.2). Unit packages shall be snugly packed in exterior containers of uniform size and shape and of minimum cube and tare.

5.2.1 Level A - Not applicable.

5.2.2 Level B - Unit packages shall be packed in fiberboard boxes, PPP-B-636, Class Domestic, Grade 275. Maximum weight and dimensions shall be in accordance with Table I of PPP-B-636. Closing and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C - Unit packages shall be packed in exterior containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet the Uniform Freight Classification Rules or the regulations of other common carriers, as applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required by the contract or order, the unit and exgerior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - Type 431 steel T-bolts are intended for use in V-band couplings and band clamps. The material should not be used in application which involve exposure to temperature in excess of 500°F for extended periods of time.

6.1.1 This process specification is intended to supplement detail design drawings and is not of itself a complete procurement document.

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6.2 Ordering information - Requisitions, orders and contracts should specify the following:

1. Number and title of this specification.
2. Configuration and dimensions of items desired by reference to applicable drawings.
3. Number of items of each size.
4 Level of preservation and packaging (5.1).
5. Level of packing (5.2).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

ECIFICATION

MIL-T-81364(WP) T-BOLTS, 180,000 PSI, TYPE 431 STAINLESS STEEL, MANUFACTURING PROCESS REQUIREMENTS

GANIZATION *(Of submitter)*

CITY AND STATE

NTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

TERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

MITTED BY *(Printed or typed name and activity)*

DATE

FORM 1426
APR 83

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